Assessment Proforma 2025-26

Key Information

Module Code	CMT307
Module Title	Applied Machine Learning
Module Leader	Yuhua Li
Module Moderator	Steven Schockaert
Assessment Title	Machine Learning Project
Assessment Number	2
Assessment Weighting	50%
Assessment Limits	A group project report of no more than 4500 words (and a maximum of 10 pages in total), a zip file with all the Python code for the group project and a README file, and a self-reflection and peer assessment using the proforma provided.

This assessment's due date and time is specified on Learning Central. To view it click on the assessment's link, which will open *'Details & Information'* section for this assessment. Alternatively, go to *'Calendar'* section of a module, and select *'Due Dates'* option. You can change the view to daily or monthly, as well as use arrows to navigate the calendar.

The COMSC Assessment Calendar for all assessments is also available on Learning Central: by going to '**COMSC-SCHOOL'** organisation -> **'Assessment & Feedback 2025 - 26'** folder -> **'COMSC Assessment Calendar 2025 - 26'** page.

Learning Outcomes

The learning outcomes for this assessment are as follows:

1. Implement and evaluate machine learning methods to solve a given task.
2. Explain the basic principles underlying common machine learning methods.
3. Choose an appropriate machine learning method and data pre-processing strategy to address the needs of a given application setting.
4. Reflect on the importance of data representation for the success of machine learning methods.
5. Critically appraise the ethical implications and societal risks associated with the deployment of machine learning methods.
6. Explain the nature, strengths and limitations of an implemented machine learning technique.

Submission Instructions

All assessment files should be submitted in '*Assessment & Feedback'* folder on Learning Central.

Before you submit your assessment, you **must complete and submit** *'Assessment Coversheet'* form. Further instructions are given in this module's 'Assessment & Feeback' folder.

This coursework submission consists of a **group submission** (note: group submission refers to the submission of the two files for Part 1 hereafter) and an **individual submission**. The group submission will be submitted via Learning Central by a nominated team member, and the individual submission will be submitted via Learning Central by individuals.

The **group submission** (from Part 1 of this assignment) consists of two files:

1. A single PDF file for your **group report** (up to 4500 words and a maximum of 10 pages in total) on a specific machine learning project.
2. A zip file containing **all source codes** of your group project.

All group members must have seen and agreed to the final version of the submission.

The **individual submission** (from Part 2 of this assignment) consists of a single PDF file for the self-reflection and peer assessment proforma.

All files should be submitted via Learning Central. The submission page can be found under 'Assessment & Feedback' in the CMT307 module on Learning Central. You must submit to Learning Central the following files using a naming convention as required in the table below, replacing [group number] with your group number and [student number] with your student ID number, e.g., *G1_Report.pdf* for the report of Group 1 and *G5_C1234567_peerAssessment*.pdf for the peer assessment proforma of student C1234567 in Group 5.

Your submission should consist of multiple files:

Description	**Compulsory or Optional?**	**Type**	**Name**
Group Coversheet	Compulsory	One PDF (.pdf) file	G[group number]_Coversheet.pdf
Group report	Compulsory	One PDF (.pdf) file	G[group number]_Report.pdf
Group Python code	Compulsory	One ZIP (.zip) file	G[group number]_Code.zip

Individual peer assessment proforma	Compulsory	One PDF (.pdf) file	G[group number] _[student number]_peerAssessment.pdf

If you are unable to submit your work, email your submission to the module leader and comsc-submissions@cardiff.ac.uk. If the technical issue is with Cardiff University systems, also report it to the University IT Support: email it-support@cardiff.ac.uk, tel. +44 (0)29 2251 1111.

Note: This coursework consists of two parts: Part 1 is for group project code and group report. Part 2 is for individual work.

Part 1: **Group report** and **project code**. The deliverable includes a zip file containing all file(s) of the source code, and a single PDF file for the written report (up to 4500 words, a maximum of 10 pages in total, minimum font size of 10, and a minimum margin of 20 mm) describing solutions, design choices, evaluation and a reflection on the main challenges faced during development and insights gained throughout the process. Before handing it in make sure all documentation has been collected. Additional supporting material, such as sources or data may also be submitted if appropriate along with the code zip file. Any code submitted will be run in Python 3 and must be submitted as stipulated in the instructions. Make sure the report clearly mentions your group number, the project title, the name of the supervisor and a list of student ID numbers of all members of the group on the title page of your report.

Part 2: **Peer assessment**. Part 2 consists of a peer assessment proforma where students reflect on individual contributions and assign marks to other members of your group. Each individual will submit the peer assessment proforma via Learning Central by the deadline. **A student who fails to submit a peer assessment proforma will be marked as "non-submission" for this group project assessment**.

Any deviation from the submission instructions above (including the number and types of files submitted) will result in a reduction in marks for this assessment by 20%.

Staff reserve the right to invite students to a meeting to discuss coursework submissions

Assessment Description

In this coursework, students demonstrate their knowledge and skills with the topics covered in the module via a group project. This coursework consists of two parts: Part 1 is for group report and project code. Part 2 is for individual work.

Marks will be awarded to the individual student based on the quality of the group report, the project code and their contribution.

Part 1: Group report (80%)

In Part 1, students will be allocated in groups to design a machine learning project on one specific topic. The list of all topics along with their descriptions is available in Appendix A.

Each group is given a specific dataset and a supervisor. The task of each group consists of developing a whole machine learning pipeline that attempts to solve the task. The usage of neural networks as methods/baselines is not mandatory but will be positively assessed; the non-usage of neural methods should be properly justified.

Throughout the course, the groups will have several milestones and should present their progress to their supervisor in each session. Finally, the group will write a report summarizing the steps followed and the main insights gained as part of the process.

As part of the group decisions, each student will be allocated to one of the following tasks:

T1- Descriptive analysis of the dataset + Result analysis
T2- Preprocessing + Literature review
T3- Implementation + Results

Each of these tasks will have a minimum of two students involved (except in exceptional cases when this is not possible), who will lead and work together in the specific task and as part of the group. All group members contribute to the delivery of the whole project. The structure of the report will be decided by the group members. In Appendix B, students can find some guidelines to write the report, including some of the common sections that groups may want to include in their report.

Note: These are just guidelines and students are not forced to follow this report structure. New sections may be added or adjusted if necessary.

Each student will also be involved in all group activities/tasks and will be responsible for the well-functioning and coordination of the team members.

Deliverables

The deliverables for this part include **a report of no more than 4500 words** (and a maximum of 10 pages in total, minimum font size of 10, and a minimum margin of 20 mm) and a **zip file with all the Python code and a README file**. The group report must have the first page from Appendix C. The code and README should contain these contents:

(1) Code to get the statistics used to complement the descriptive analysis of the dataset.
(2) Code to train models on the training set and evaluate them on the test set. This code should also include all steps for preprocessing the original dataset if it were necessary.
(3) A README file explaining how to run the code for each of the two parts.

The code will not be marked separately and will only be used as a complement to assess specific parts of the report.

Assessment

The final mark for this part (80% of the total marks) will result from the following items:

- T1- Descriptive analysis of the dataset + result analysis (20%)
- T2- Preprocessing + Literature review (20%)
- T3- Implementation + Results (20%)
- T4- Group report as a whole, including its coherence and structure (20%)

Note: Normally every member of the group will receive the same mark for the group report and project code, but in some cases, marks might be weighted by the individual contribution to the project. This would be based on peer assessment for which instructions will be given in Part 2.

Part 2: Self-reflection and peer assessment (20%)

In Part 2, students are asked to do a self-reflection and peer assessment using the proforma in Appendix D. In the proforma, you must discuss your and each member's contribution to the group project and to the overall group report. You must show that you contributed to the group report and code. Discuss what tasks you have performed and provide evidence of your work (you may refer to the group report and code for the actual work/results). Discuss how you approached these tasks and how you interacted with other members, both in sharing your results and in organising the team's activities. Consider how well your existing skills were utilised and what new skills you have learned. Then reflect on your overall performance and role in the team and suggest what went well and what changes you will be making to improve (1) your performance in particular, and (2) the performance and results of methods and analyses performed as part of the project. You may also reflect on how your perspective and approach changed over time and adapted to improve your work.

Note: Please indicate the information about your group (group number, project name) in the proforma.

This part weighs 20% of the total marks.

Contribution of group members

This is a team project and this assignment is assessed as a team, apart from the individual mark for the supporting evidence for each team member's service. Each team member is expected to contribute to the project for the tasks that are agreed upon in the group.

You will also be asked to submit a peer assessment form. You will evaluate the contribution of each group member and your own contribution to the deliverables. Normally every member of the group will receive the same mark for the group components except in the case where a group member's contribution and/or quality of work falls significantly below that of the majority of the group, in which case the marks will be adjusted accordingly. **A student who does not contribute at all to their group will receive zero marks.**

Please inform the module leader if there are any problems with any group members not engaging in group tasks or missing group meetings. The teaching team will check on the engagement of the group members in the contact sessions and review meetings. Students should therefore inform the module leader (and the other team members, if appropriate) if circumstances arise that are likely to affect their engagement with their work and/or attendance at weekly meetings with the rest of the team.

The teaching team will provide formative feedback during contact sessions. Your team should also meet regularly outside of these.

Assessment Criteria

Group report and code (total 80%)

T1-1. Descriptive analysis of the dataset (10%)

High Distinction 80%+	Thorough and insightful data exploration.
Distinction 70-79%	Extensive and informative data exploration.
Merit 60-69%	Good data exploration but misses some insightful analysis.
Pass 50-59%	Suitable but limited data exploration.
Marginal Fail 40-49%	Little and arbitrary data exploration.
Fail 0-39%	No or meaningless data exploration.

T1-2. Result analysis (10%)

High Distinction 80%+	Thorough and critically insightful result analysis and discussion.
Distinction 70-79%	Comprehensive and insightful result analysis and discussion.
Merit 60-69%	Good result analysis and discussion but lack of depth.
Pass 50-59%	General result analysis and discussion.
Marginal Fail 40-49%	Little result analysis and discussion.
Fail 0-39%	No or little meaningful result analysis and discussion.

T2-1. Preprocessing (10%)

High Distinction 80%+	Innovation and extensive pre-processing to deal with all aspects of non-ideal characteristics of the data with an aim to achieve the best performance
Distinction 70-79%	Extensive pre-processing to deal with major aspects of non-ideal characteristics of the data.
Merit 60-69%	Adequate pre-processing to prepare the data for model development.
Pass 50-59%	Some necessary pre-processing is conducted.
Marginal Fail 40-49%	Some basic but omitted necessary pre-processing.
Fail 0-39%	No or very little data pre-processing

T2-2. Literature review (10%)

High Distinction 80%+	Exceptionally well articulated and critical literature review
Distinction 70-79%	Extensive and critical literature review.
Merit 60-69%	Adequate literature review.
Pass 50-59%	Basic and general literature review.
Marginal Fail 40-49%	Superficial literature review, missed major related work.
Fail 0-39%	No or very little literature review.

T3-1. Implementation (15%)

High Distinction 80%+	The implementation provides a considerably original, innovative and superior solution to the problem.
Distinction 70-79%	All models are excellently implemented, properly trained and systematically optimised.
Merit 60-69%	The implemented models are properly trained and optimised.
Pass 50-59%	The models are implemented but not properly/ sufficiently trained.
Marginal Fail 40-49%	Unsuitable ML method is chosen. The models are not correctly implemented and optimised.
Fail 0-39%	Superficial demonstration of only basic and probably non-working implementation.

T3-2. Results (5%)

High Distinction 80%+	Thorough model evaluation for accurate and reliable generalisation.
Distinction 70-79%	Comprehensive model evaluation for best results.
Merit 60-69%	Good performance evaluated with suitable metrics.
Pass 50-59%	Performance evaluation using metrics without considering data characteristics.
Marginal Fail 40-49%	Little and arbitrary performance evaluation.
Fail 0-39%	No or improper performance evaluation.

T4. Group report as a whole, including its coherence and structure (20%)

High Distinction 80%+	Professionally, concisely and cohesively presented to a publishable standard. Insightful discussion and accurately drawn conclusions.
Distinction 70-79%	Very well organised and clearly written with good technical content. Results assessed critically and arguments very well presented and supported.
Merit 60-69%	Well organised and clearly written with sound technical content. Results analysed and assessed well.
Pass 50-59%	The report is acceptable in terms of technical contents and structure. General discussion and vague conclusion.
Marginal Fail 40-49%	Poor style of writing with some parts difficult to follow. Poor organisation and presentation of material.
Fail 0-39%	Difficult to read and lacks a logical train of thought or argument. Very poor organisation and presentation of material.

Help and Support

Questions about the assessment can be asked at the scheduled meetings/classes with your group supervisor during the group project period.

Feedback

Feedback on your coursework will address the above assessment criteria. Feedback and marks will be returned via Learning Central in accordance with the university's Academic Feedback Policy. You can see feedback return dates for each assessment on Learning Central: see **'COMSC Assessment Calendar 2025 - 26'** page.

Appendix A: CMT307 Group Projects

Note 1: Datasets are provided for all projects

Note 2: Not all datasets contain train/dev/test splits. It is up to the group members to decide on a suitable split in those cases (or cross-validation).

Note 3: All links to datasets were checked and working at the time when we tested. If a dataset is no longer accessible from the given links, please let us know, and we will provide alternative data access.

1. **Emoji prediction**

The goal of this task is to predict an emoji (e.g. 🤔) given a tweet. Dataset based on the SemEval 2018 task on emoji prediction (https://competitions.codalab.org/competitions/17344).

Direct link to download the dataset: emoji_prediction.zip

2. **Text categorization**

Text categorization (also referred to as text classification) consists of associating a document with a given topic (e.g. sports, politics, etc.). 20 Newsgroups dataset (http://qwone.com/~jason/20Newsgroups/).

Direct link for downloading the dataset (by date version): http://qwone.com/~jason/20Newsgroups/20news-bydate.tar.gz

3. **Hate speech detection**

Given a tweet or a piece of comment, the task of hate speech consists of predicting whether the given text represents hate speech or not and classifying it accordingly. This task is based on the SemEval 2019 task on the detection of hate speech against immigrants and women on Twitter (https://competitions.codalab.org/competitions/19935).

Direct link to download the dataset:https://drive.google.com/file/d/1Cn60H0klYNRNI_q5SeghzFlu6eMYmMOO/view?usp=sharing

4. **Opinion vs. factual news stories**

When building models to analyse news data at scale, it is important to distinguish between articles where the author is reporting "facts" (e.g. that something happened, or that a person said something) and those where the author is reporting their own opinion (e.g. that they think something will happen or that something happening is a good thing). Using a dataset of financial news articles, the task consists of building a classifier which labels an article as being fact or opinion. (Note: this dataset may contain noisy labels as they were obtained semi-automatically).

Direct link to download the dataset (dataset provided by AYLIEN):
https://drive.google.com/file/d/1Mqoh7gG-g3Sc3Zh6zEO8o2sKBpdSeFiJ/view?usp=sharing

5. **Fine-grained image classification**

The goal of this task is to develop an algorithm to learn to classify images containing objects of the same category (e.g. birds, dogs) into specific sub-categories, i.e. specific species. Datasets are available:

Caltech-UCSD Birds-200-2011: 200 categories of birds
https://paperswithcode.com/dataset/cub-200-2011
(http://www.vision.caltech.edu/visipedia/CUB-200-2011.html)

Stanford dogs: 120 categories of dogs

http://vision.stanford.edu/aditya86/ImageNetDogs/

You can choose to use either birds or dogs dataset.

6. **Detection and recognition of traffic signs**

Detection and recognition of traffic signs is an important task for autonomous driving. The task is to identify traffic signs in images and recognise them.

Dataset available at https://www.kaggle.com/datasets/meowmeowmeowmeowmeow/gtsrb-german-traffic-sign,
(https://sid.erda.dk/public/archives/daaeac0d7ce1152aea9b61d9f1e19370/published-archive.html) (German Traffic Sign Recognition Benchmark)

7. **Object localisation**

It is straightforward for people to locate objects in an image, but can you develop a system to learn to do this? Given an image, the task is to find all the instances of relevant objects (as bounding boxes).

Dataset available at http://host.robots.ox.ac.uk/pascal/VOC/voc2012/index.html

It contains multiple datasets, and you have the flexibility to choose the ones that satisfy your project needs. So the following are just suggestions, and you may come up with your own approach, as long as it is reasonable and well justified.

For object detection/localisation, the most relevant dataset is the one used for

> Detection: Predict the bounding box and label of each object from the twenty target classes in the test image.

You can see example images for this task at http://host.robots.ox.ac.uk/pascal/VOC/voc2012/examples/index.html

8. **Urban Sound Classification**

Dataset from here: https://urbansounddataset.weebly.com/urbansound8k.html . The goal of this task is to classify different urban sounds (e.g. dog bark or street horn) into their correct classes.

Direct link to download the dataset: https://drive.google.com/file/d/15fojQ3xKcPMLwIm6s8hoyM0xJjZ6SOZU/view?usp=sharing

Note: the size of this dataset is over 6GB (a portion of it may be used if hard to process). Therefore an important part of this project is the preprocessing and handling of the data. The following repository includes a machine learning model that can be used as a guide/reference at the beginning: https://github.com/AmritK10/Urban-Sound-Classification

9. **Energy usage prediction**

Prediction of building energy consumption is important to electricity distribution, management and the environment. This project will develop a machine learning model to predict energy usage based on historic usage rates and observed weather. The dataset is available at https://www.kaggle.com/c/ashrae-energy-prediction/overview.

10. **Stock price prediction**

Machine learning can be useful for algorithmic trading. It can help to develop a profitable trading strategy by predicting a stock price in the future based on historical daily OHLC price data. Data can be readily downloaded from Yahoo Finance using the Python package yfinance: https://pypi.org/project/yfinance/.

11. **Sound-based lung disease detection**

Respiratory health has a significant impact on society and the economy. Globally, respiratory diseases were the third leading cause of death. This project will apply machine learning techniques to assess respiratory conditions based on audio recordings from individuals. It will help develop scalable and accessible respiratory diagnostics via a phone call.

Respiratory Sound Database: https://www.kaggle.com/datasets/vbookshelf/respiratory-sound-database.

Background: https://bhichallenge.med.auth.gr/ICBHI_2017_Challenge

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For groups working on image-related tasks (Projects 5-7):

Some tutorials for basic image processing with Keras/Tensorflow:

https://stackabuse.com/image-recognition-in-python-with-tensorflow-and-keras/

https://developer.ibm.com/technologies/artificial-intelligence/articles/image-recognition-challenge-with-tensorflow-and-keras-pt1

https://www.tensorflow.org/tutorials/keras/classification

Appendix B: Group report guidelines

In this document, we detail some of the common sections involved in a machine learning project report. These sections are only presented as a guideline, but the report may have a different structure both in terms of sections and order.

1. **Introduction**

Summary of the task and main goals/contributions/insights of the project.

2. **Literature review / Related work**

Overview of the related work most connected to the methods and tasks of the projects. Explain the differences and the connection between works in the literature with respect to the employed method (e.g. advantages/disadvantages, ideas you exploited, etc.).

Tip: Google Scholar is a good resource to find relevant articles on any of the topics.

3. **Description of the task/dataset**

Description of the task and dataset, including relevant statistics of dataset splits.

4. **Methodology**

Description of the machine learning methods used in the project.

5. **Experimental setting**

Description of the specific details of the evaluation (e.g. parameter tuning, usage of the development set).

6. **Results**

Final results of the experiments, including baselines and table/s with precision/recall/accuracy/f1, etc.

7. **Analysis**

Analysis of the results, error analysis (investigate the type of error the system makes, etc.).

8. **Conclusion and future work**

Summary of the main conclusions and takeaways from the experiments. Explain ways to investigate or improve the method in the future.

References

List references in IEEE reference style. See the IEEE Reference Guide for how to cite a variety of references in IEEE style.

Appendix C: Group report front page

You will include the following information on the first page of your group report.

CMT307 Coursework 2 Group Project

Group number	
Project title	
Supervisor	
Word count	[give the number of words in your report]
	[list student IDs of all group members here]

Appendix D: Peer Assessment and Self-reflection

Notes for completing the peer assessment form

1. Each row of the form is for one member
2. In the first column, add the full names of all other group members (excluding yourself) in alphabetical order by surname.
3. In the second column, evaluate the contributions of each member of the group to the project (up to 150 words for each member). This evaluation should be based on evidence, such as tasks completed, interaction with other members, participation in group activities, team working skills, etc.
4. In the third column, fairly and honestly allocate marks to each member based on your evaluation in the second column. **You have a total of 20 marks to be allocated to all other members**. For example, for a group with 4 other members, an allocation of 20 marks to those 4 members could be 7, 3, 10, 0, respectively. A 0 mark indicates that the member has done nothing to the group project. If contributions are felt to be fair, you can score the same marks to all other students in your group.
5. Total peer assessment (TPA) marks of a member will be obtained by aggregating marks of all peer assessment forms in the group, and a member can have a maximum peer assessment mark of 20 (even if her/his aggregated TPA mark is greater than 20).
6. The final peer assessment (FPA) marks of a member will be weighted by the total group mark from Part 1. For example, if the total group mark is 57 out of 80 marks and a member called Boris has a TPA mark of 16, then Boris's FPA will be (57/80) x 16 = 11 (round to nearest integer). Finally, Boris's overall mark for this coursework will be 57+11=68%.
7. If you do not submit the Peer Assessment and Self-reflection Proforma, you will be marked as "*non-submission*". If the submitted proforma is considered unusable (for example, if the mark allocation is clearly unfair), your FPA will be zero.
8. Each member of the group will typically receive the same marks for the group component. However, if a member's contribution or the quality of their work is significantly lower than that of the majority, their grade will be adjusted downwards. Additionally, **a student who does not contribute at all to the group will receive a mark of zero**.

Peer Assessment and Self-reflection Proforma

Student ID:

Group number:

Project title:

Supervisor:

Peer assessment

Member name	Contribution and justification	Marks

Self-reflection

[write your self-reflection here, up to 300 words. Your self-reflection will be used to cross-check if you have been fairly assessed by your peers]